



Formal Methods: Multiplying Decimals

Solving Problems with Multiplication and Division Step 8

Small Step: Use formal methods to multiply decimals.

1. Given that $47 \times 6 = 282$, calculate each of these multiplications.

a. 47×0.6

b. 4.7×6

c. 4.7×0.6

d. 0.47×0.6

2. Calculate each of the following.

a. 0.3×0.5

b. 0.7×0.2

c. 0.2×0.9

d. 0.04×20

e. 0.5×300

f. 0.7×0.8

g. 0.4×3

h. 0.6×50



3. Samir and Lucia are asked to calculate 23.7×1.8 .
- a. Samir uses the column method. He starts by estimating 23.7×1.8 by rounding each number to 1 significant figure. What answer will Samir get?

- b. Samir then calculates 237×18 . Complete his working below.

$$\begin{array}{r} \\ \times \\ \hline \end{array}$$

- c. What is the correct answer to 23.7×1.8 ?
- _____
- d. Lucia is also asked to calculate 23.7×1.8 . She sets up a grid as shown below. Complete Lucia's method.

$\times$	20	3	0.7
1			
0.8			

4. Estimate each of the calculations below.

a. 3.7×29

b. 4.1×87

c. 8.9×7.3

d. 23.7×0.95

5. Use your answers to question 4 to help you find the exact answer to each of these multiplications.

a. 3.7×29

b. 4.1×87



c. 8.9×7.3

d. 23.7×0.95

6. Use a method of your choice to calculate each of the following.

a. 7.5×3.1

b. 8.6×4.9

c. 13.6×5.3

d. 0.37×4.2

e. 7.84×62.9

f. 1.79×0.41

g. 8.6×58.5

h. 0.759×0.62

7. Kay buys 4.2m of ribbon. The ribbon costs £3.20 per metre. How much does Kay spend in total?



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8. Joey went to the store and bought 2.5kg of potatoes for £1.24 per kilogram. They also purchased 1.75kg of carrots for 60p per kilogram. How much did Joey spend in total on potatoes and carrots?

Challenge

In the pyramids below, each brick is the product of the two bricks below. Complete the pyramids.

